

Interior Hölder estimates for kinetic equations with measurable diffusion

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Abstract

We prove an interior kinetic Hölder estimate for bounded intrinsic classical solutions of $Yu = \text{tr}(AD_v^2 u)$, where the symmetric diffusion matrix is Borel measurable and uniformly elliptic, with arbitrary dependence on time, position, and velocity. The central estimate is a uniform spacetime $L^{1+\varepsilon}$ occupation bound for all adapted kinetic diffusions, with $\varepsilon = \lambda^2/(256d^2\Lambda^2)$. Its proof uses ordinary phase Gaussian convolution with a negative position–velocity cross covariance. A controlled-path argument and a self-contained kinetic covering proof turn this bound into oscillation decay. A common equation-residual approximation preserves the original intrinsic solution class and the closed top time face.

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1 Statement and proof structure

Fix $d \geq 1$ and $0 < \lambda \leq \Lambda < \infty$. For $z = (t, x, v) \in \mathbb{R}^{1+2d}$ write

$$Y = \partial_t + v \cdot \nabla_x.$$

For $z_0 = (t_0, x_0, v_0)$ and $r > 0$, let

$$Q_r(z_0) = \left\{ (t, x, v) : \begin{array}{l} -r^2 < t - t_0 \leq 0, \\ |x - x_0 - (t - t_0)v_0| < r^3, \quad |v - v_0| < r \end{array} \right\}.$$

The notation $Q_r^\circ(z_0)$ replaces the upper time inequality by $t - t_0 < 0$. All vector norms are Euclidean. Define

$$d_{\text{kin}}(z_1, z_2) = \min_{w \in \mathbb{R}^d} \max \{ |t_1 - t_2|^{1/2}, |v_1 - w|, |v_2 - w|, 2^{-1/3} |x_1 - x_2 - (t_1 - t_2)w|^{1/3} \}.$$

For $0 < \alpha < 1$ and a set U , put

$$[u]_{C_{\text{kin}}^\alpha(U)} = \sup_{\substack{z, z' \in U \\ z \neq z'}} \frac{|u(z) - u(z')|}{d_{\text{kin}}(z, z')^\alpha}, \quad \text{osc}_U u = \sup_U u - \inf_U u.$$

Theorem 1.1. *There exist $\alpha = \alpha(d, \lambda, \Lambda) \in (0, 1)$ and $C = C(d, \lambda, \Lambda) < \infty$ with the following property. Let $r > 0$, let $z_0 \in \mathbb{R}^{1+2d}$, and let $A : Q_r^\circ(z_0) \rightarrow \text{Sym}_d$ be Borel measurable with*

$$\lambda |\xi|^2 \leq \xi^\top A(z) \xi \leq \Lambda |\xi|^2 \quad \text{for every } z \in Q_r^\circ(z_0), \xi \in \mathbb{R}^d.$$

Suppose u is bounded and continuous on $Q_r(z_0)$, that Yu and $D_v^2 u$ exist and are continuous on $Q_r^\circ(z_0)$, and that

$$Yu = \text{tr}(AD_v^2 u) \quad \text{pointwise on } Q_r^\circ(z_0).$$

Then

$$r^\alpha [u]_{C_{\text{kin}}^\alpha(Q_{r/2}(z_0))} \leq C \text{osc}_{Q_r(z_0)} u.$$

No separate existence or continuity of $\partial_t u$ or $D_x u$ is assumed. Neither coefficient continuity nor coefficient derivative bounds are required.

We prove all analytic and kinetic geometric estimates used below. The standard background is elementary stochastic calculus: existence of smooth Lipschitz-coefficient stochastic differential equations, Itô's formula, the Doob and Burkholder–Davis–Gundy martingale inequalities, bounded-integrand Girsanov change of measure, and optional sampling for bounded stopped supermartingales. We also use Gaussian integration by parts and weak compactness in L^p , $1 < p < \infty$. No kinetic Harnack or Hölder theorem is an input. In particular, the covering argument below includes its differentiation and stack-union ingredients.

The forward-stack geometric mechanism has precedents in the kinetic covering argument of Imbert and Silvestre [1, Section 10]. We give its proof here, including a boundary-trimming argument; no theorem from that work is imported.

The occupation theorem concerns actual processes and therefore assumes no martingale-problem existence theorem for Borel fields. In the proof of the main result, processes are used only for globally smooth coefficients. Borel coefficients are handled by an approximation of the equation residual on compact subsets of the original domain.

2 A uniform occupation estimate

We first prove a quantitative estimate for actual adapted diffusions. In particular, this estimate does not require a Markov property.

Theorem 2.1 (Occupation estimate). *Let $T > 0$ and let a continuous adapted process satisfy*

$$dX_t = V_t dt, \quad dV_t = \sqrt{2A_t} dW_t, \quad \lambda I \leq A_t \leq \Lambda I,$$

where A_t is progressively measurable and symmetric. Assume that its initial law has compact support. Set

$$e = \frac{\lambda^2}{256d^2\Lambda^2}, \quad q = 1 + e, \quad a = \frac{\lambda}{2}, \quad B = (\pi\sqrt{3}a)^{-de}.$$

The spacetime occupation measure has a density p and

$$\int_0^T \int_{\mathbb{R}^{2d}} p^q dx dv dt \leq B \left(2T + \frac{8}{1-2de} \right) =: C_q. \quad (1)$$

Consequently, for every spacetime Borel set E ,

$$\mathbb{E} \int_0^T \mathbf{1}_E(t, X_t, V_t) dt \leq C_q^{1/q} |E|^{(q-1)/q}. \quad (2)$$

The same estimate holds for occupation stopped at any stopping time at most T . Its constants do not depend on the initial compact support.

Proof. Write $y = (x, v)$, let m_t be the phase law, and define the matrix measure $n_t(E) = \mathbb{E}[A_t \mathbf{1}_{\{(X_t, V_t) \in E\}}]$ for almost every t . The forward identity and the matrix inequalities are

$$Ym = D_v^2 : n, \quad \lambda mI \leq n \leq \Lambda mI.$$

For the auxiliary parameter $h > 0$, introduce an ordinary phase Gaussian k_h with covariance

$$\Sigma_h = a \begin{pmatrix} h^3 I & -h^2 I/2 \\ -h^2 I/2 & hI \end{pmatrix}, \quad K_h = \Sigma'_h = a \begin{pmatrix} 3h^2 I & -hI \\ -hI & I \end{pmatrix}. \quad (3)$$

Both matrices are positive definite. Define, at each fixed physical time,

$$p_h = m_t * k_h, \quad N_h = n_t * k_h, \quad C_h = N_h / p_h.$$

Here $p_h > 0$ and $\lambda I \leq C_h \leq \Lambda I$. The convolutions are ordinary convolutions in (x, v) , not convolutions on the kinetic group. Their auxiliary equations are

$$\partial_h p_h = \frac{1}{2} K_h : D_y^2 p_h, \quad \partial_h N_h = \frac{1}{2} K_h : D_y^2 N_h. \quad (4)$$

The physical equation gains a useful full phase diffusion. Direct convolution of transport gives

$$Yp_h = D_v^2 : N_h - (\Sigma_h)_{vx} : D_x^2 p_h - (\Sigma_h)_{vv} : D_x D_v p_h.$$

For a sign check, add an independent static Gaussian (R, S) with covariance Σ_h to (X_t, V_t) . In the new coordinates, the position drift is the displayed velocity minus S . Gaussian integration by parts gives

$$\mathbb{E}[S \delta_{(X_t+R, V_t+S)}] = -(\Sigma_h)_{vx} D_x p_h - (\Sigma_h)_{vv} D_v p_h,$$

which yields precisely the preceding identity. Thus

$$Yp_h = D_y^2 : (D_h p_h), \quad D_h = \begin{pmatrix} ah^2 I/2 & -ahI/2 \\ -ahI/2 & C_h \end{pmatrix}, \quad \text{div}_y D_h = (0, \text{div}_v C_h). \quad (5)$$

The mixed block occurs twice in the double divergence. Two elementary matrix inequalities are

$$D_h \geq \frac{1}{8} K_h + \frac{\lambda}{4} P_v, \quad K_h \geq \frac{\lambda}{3} P_v, \quad (6)$$

where P_v projects onto velocity. For the first, normalize the x coordinate by h . The difference is bounded below by

$$a \begin{pmatrix} I/8 & -3I/8 \\ -3I/8 & 11I/8 \end{pmatrix} > 0,$$

whose scalar block determinant is $a^2/32$. The second follows from the velocity Schur complement $2a/3$ of K_h .

All the next integrals are in y . Write $s = D_y \log p_h$ and set

$$\begin{aligned} H &= \int p_h^q, & F &= \int p_h^q |C_h|_F^2, & I_q &= \int p_h^q s^\top K_h s, \\ J_q &= \sum_{i,j} \int p_h^q (D_y C_{h,ij})^\top K_h D_y C_{h,ij}, & J_{q,v} &= \sum_{i,j} \int p_h^q |D_v C_{h,ij}|^2. \end{aligned}$$

Equation (4) yields

$$H_h = -\frac{1}{2}q(q-1)I_q. \quad (7)$$

The quotient obeys

$$\partial_h C_h = \frac{1}{2}K_h : D_y^2 C_h + (K_h s) \cdot D_y C_h.$$

Differentiating F entry by entry and integrating by parts gives

$$\begin{aligned} -F_h &= \int p_h^q \left(|D_y C_h|_{K_h}^2 + 2(q-1) \sum_{i,j} C_{h,ij} (D_y C_{h,ij})^\top K_h s \right. \\ &\quad \left. + \frac{1}{2}q(q-1) |C_h|_F^2 s^\top K_h s \right). \end{aligned} \quad (8)$$

For $1 < q < 2$, put $U = |D_y C_h|_{K_h}$ and $W = |C_h|_F |s|_{K_h}$. The inequality

$$2(q-1)UW \leq \frac{2(q-1)}{q}U^2 + \frac{q(q-1)}{2}W^2$$

shows that

$$-F_h \geq \frac{2-q}{q}J_q, \quad 0 \leq F \leq d\Lambda^2 H. \quad (9)$$

Testing (5) by qp_h^{q-1} gives

$$H_t = -q(q-1) \int p_h^q s^\top D_h s - q(q-1) \int p_h^q (\operatorname{div}_v C_h) \cdot D_v \log p_h.$$

Transport integrates to zero. Since $|\operatorname{div}_v C_h|^2 \leq d|D_v C_h|_F^2$, Young's inequality uses exactly the velocity reserve in (6) and gives

$$H_t + \frac{1}{8}q(q-1)I_q \leq \frac{dq(q-1)}{\lambda}J_{q,v} \leq \frac{3dq(q-1)}{\lambda^2}J_q.$$

Together with (7) and (9), this is

$$H_t - \frac{1}{4}H_h \leq -k_q F_h, \quad k_q = \frac{3dq^2(q-1)}{\lambda^2(2-q)}. \quad (10)$$

Integrate over $\eta < h < 1$ and $0 < t < T$. Discarding the nonnegative subtracted terms $H(h, T)$ and $F(1, t)$, and using $F \leq d\Lambda^2 H$, we obtain

$$\left(\frac{1}{4} - k_q d\Lambda^2\right) \int_0^T H(\eta, t) dt \leq \frac{1}{4} \int_0^T H(1, t) dt + \int_0^1 H(h, 0) dh. \quad (11)$$

For the displayed e , one has $e \leq 1/256$, $q < 3/2$, and

$$k_q d\Lambda^2 \leq \frac{27}{2} \frac{d^2 \Lambda^2}{\lambda^2} e = \frac{27}{512} < \frac{1}{8}.$$

Also

$$\det \Sigma_h = (3a^2 h^4 / 4)^d, \quad \|k_h\|_\infty = (\pi \sqrt{3} a)^{-d} h^{-2d}.$$

Since p_h is a probability density bounded by this supremum,

$$H(h, t) \leq \|k_h\|_\infty^{q-1} = B h^{-2de}.$$

This holds even at $t = 0$ for an atomic initial law. Because $2de < 1$, (11) proves the uniform bound

$$\int_0^T H(\eta, t) dt \leq B \left(2T + \frac{8}{1 - 2de} \right).$$

We record the regularization details, since physical-time coefficients are only measurable. On any compact auxiliary interval $\eta \leq h \leq 1$, bounded volatility and compact initial support give uniform polynomial moments of every order. Gaussian differentiation therefore bounds p_h, N_h and their derivatives in polynomially weighted L^1 . Conditional Jensen applied to the Gaussian score gives

$$\int p_h |D_y \log p_h|^2 \leq \text{tr}(\Sigma_h^{-1}).$$

Differentiating the conditional mean of the bounded mark A_t similarly bounds $\int p_h |D_y C_h|_F^2$; weighted versions follow by the same argument with the phase and Gaussian moments. The factor p_h^{q-1} is uniformly bounded for this auxiliary interval. These estimates justify all Fisher and cross terms above, including the negative powers of p_h in intermediate quotient formulas.

First perform the calculations with phase cutoffs on balls of radii L and $2L$; all boundary terms tend to zero by these weighted estimates. Convolution of m, n in physical time on interior intervals commutes with Y and preserves their cone relation. It justifies the time chain rules and passes to the integrated inequality. At fixed η , weak continuity of the actual phase law gives local continuity of p_h , and $p_h^q \leq \|k_h\|_\infty^{q-1} p_h$ controls endpoint tails uniformly. Thus the physical endpoints in the integrated inequality are valid. No endpoint density for the actual law is assumed.

The uniform bound gives weak compactness of p_η in $L^q((0, T) \times \mathbb{R}^{2d})$. On compact continuous tests, ordinary Gaussian approximation identifies the measure limit with the actual occupation measure. Weak lower semicontinuity proves (1). Hölder's inequality proves (2). A stopped occupation measure is a positive submeasure and inherits the same bound. $\square$

Reflection in x gives the same result for $dX = -V dt$. On a kinetic cylinder of radius r , translation and scaling therefore give the occupation modulus

$$\mathbb{E} \int \mathbf{1}_E(t, Z_t) dt \leq r^2 \omega(r^{-(4d+2)} |E|), \quad (12)$$

with one structural modulus on each fixed normalized time horizon. Only one sufficiently small value of this modulus will be used.

3 Uniform probabilities of following a kinetic path

Lemma 3.1 (Controlled tube). *Fix $0 < h_0 \leq H_0$, $M_0 < \infty$, and $\rho > 0$. Let $H \in [h_0, H_0]$ and let $\gamma = (\gamma_x, \gamma_v)$ be a deterministic path with $\gamma'_x = -\gamma_v$ and $|\gamma'_v| \leq M_0$. For any actual adapted process starting at $\gamma(0)$ and satisfying*

$$dX_s = -V_s ds, \quad dV_s = \sigma_s dW_s, \quad 2\lambda I \leq \sigma_s \sigma_s^\top \leq 2\Lambda I,$$

there is a structural $p_ > 0$ such that*

$$\mathbb{P} \left(\sup_{s \leq H} |X_s - \gamma_x(s)| < \rho, \sup_{s \leq H} |V_s - \gamma_v(s)| < \rho \right) \geq p_*.$$

Here p_ depends only on $d, \lambda, \Lambda, h_0, H_0, M_0, \rho$. No derivative of the diffusion coefficient enters this bound.*

Proof. Set

$$\eta = \min(\rho/8, 1), \quad \delta_0 = \min \left(h_0, 1, \frac{\eta^2}{16d\Lambda} \right), \quad N = \lceil H/\delta_0 \rceil, \quad \Delta = H/N.$$

Then $\delta_0/2 \leq \Delta \leq \delta_0$ and $N \leq \lceil H/\delta_0 \rceil$. At $t_j = j\Delta$, let e_j, f_j be the position and velocity errors relative to γ . Clip these errors radially to the balls of radii $\Delta\eta$ and η , and call the results $\bar{e}_j, \bar{f}_j$. Define

$$\begin{aligned} H_{00}(r) &= 2r^3 - 3r^2 + 1, & H_{10}(r) &= r^3 - 2r^2 + r, \\ b_j(s) &= \bar{e}_j H_{00}(s/\Delta) - \Delta \bar{f}_j H_{10}(s/\Delta), & 0 \leq s \leq \Delta, \end{aligned}$$

and prescribe the bounded adapted control

$$u_{t_j+s} = \gamma'_v(t_j + s) - b_j''(s).$$

This is defined on every path, including paths on which tracking fails. The Hermite identities give

$$b_j(0) = \bar{e}_j, \quad b_j'(0) = -\bar{f}_j, \quad b_j(\Delta) = b_j'(\Delta) = 0.$$

On $[0, 1]$, the respective bounds on H_{00}, H_{10} are 1, 1; those on their first derivatives are 3/2, 1; and those on their second derivatives are 6, 4. Consequently

$$|b_j| \leq 2\Delta\eta, \quad |b_j'| \leq 5\eta/2, \quad |b_j''| \leq 10\eta/\Delta \leq 20\eta/\delta_0.$$

Thus $|u_s| \leq M := M_0 + 20\eta/\delta_0$ everywhere.

Put $\theta_s = \sigma_s^{-1} u_s$. Then $|\theta_s| \leq M/\sqrt{2\lambda}$. The bounded-integrand exponential

$$L = \exp \left(\int_0^H \theta_s dW_s - \frac{1}{2} \int_0^H |\theta_s|^2 ds \right)$$

is a martingale density. Under $d\mathbb{Q} = L d\mathbb{P}$, $W^\mathbb{Q} = W - \int \theta ds$ is Brownian, and the same process obeys

$$dX = -V ds, \quad dV = u ds + \sigma dW^\mathbb{Q}.$$

For each interval let

$$M_j(s) = \int_{t_j}^{t_j+s} \sigma_r dW_r^\mathbb{Q}, \quad G_j = \left\{ \sup_{s \leq \Delta} |M_j(s)| \leq \eta \right\}.$$

Conditional Doob and the bracket bound give

$$\mathbb{E}_{\mathbb{Q}}[\sup |M_j|^2 \mid \mathcal{F}_{t_j}] \leq 8d\Lambda\Delta \leq \eta^2/2.$$

Thus $\mathbb{Q}(G_j \mid \mathcal{F}_{t_j}) \geq 1/2$ and $\mathbb{Q}(G) \geq 2^{-N}$ for $G = \bigcap_j G_j$. This induction uses adaptedness, not independence of the coefficients on the intervals.

On G , clipping never changes the data. Indeed, starting with zero errors and assuming the stated clipped bounds at t_j , the exact error formulas are

$$V_{t_j+s} - \gamma_v(t_j + s) = -b'_j(s) + M_j(s),$$

$$X_{t_j+s} - \gamma_x(t_j + s) = b_j(s) - \int_0^s M_j(r) dr.$$

Their suprema are at most $7\eta/2$ and $3\Delta\eta$, respectively; the endpoint errors are at most η and $\Delta\eta$. This closes the induction and makes both full-path errors strictly smaller than ρ .

The exponential identity with integrand 2θ gives

$$\mathbb{E}_{\mathbb{P}} L^2 \leq \exp(M^2 H_0 / (2\lambda)).$$

Cauchy–Schwarz therefore shows that

$$\mathbb{P}(G) \geq 2^{-2\lceil H_0/\delta_0 \rceil} \exp(-M^2 H_0 / (2\lambda)) > 0.$$

This proves the lemma. $\square$

Given phase endpoints in fixed compact sets and $H \in [h_0, H_0]$, take γ_x to be the cubic Hermite polynomial with its prescribed endpoint positions and endpoint derivatives equal to the negatives of the prescribed velocities. Set $\gamma_v = -\gamma'_x$. Its acceleration and its entire phase range are uniformly bounded in terms of those compact sets and the time interval. Thus a fixed larger phase box contains all these paths and their fixed small tubes. When a prescribed domain is used, we will explicitly ensure that the chosen paths have this margin inside it.

4 From occupation integrability to oscillation reduction

Throughout this section the ellipticity triple (d, λ, Λ) is fixed. Put $D_{\text{hom}} = 4d + 2$ and write $Q_1 = Q_1(0, 0, 0)$. We use Theorem 2.1 in the following form: for every fixed compact set of starting states and fixed time horizon, there is a common modulus $\omega(a) \rightarrow 0$ as $a \downarrow 0$ such that

$$\mathbb{E} \int_0^H \mathbf{1}_F(T_s, X_s, V_s) ds \leq \omega(|F|). \quad (13)$$

Here the backward diffusion satisfies

$$T_s = t - s, \quad dX_s = -V_s ds, \quad dV_s = \sqrt{2A(T_s, X_s, V_s)} dW_s, \quad (14)$$

and the coefficient is smooth, globally extended, and in the fixed ellipticity interval. Reflection of time and position converts this process to the forward convention of the occupation theorem. Stopped occupation satisfies the same bound. Kinetic translation and dilation multiply the bound by r^2 and the volume argument by $r^{-D_{\text{hom}}}$.

We also use Lemma 3.1: over durations in $[h_0, H_0] \subset (0, \infty)$, any deterministic kinetic path $\gamma'_x = -\gamma_v$ with a common acceleration bound can be followed within any fixed positive phase margin

with a uniform positive probability. This assertion concerns the same process (14). Cubic Hermite interpolation joins endpoints in fixed phase compacts, with uniform acceleration and a common bounded phase range. All constants in this section are independent of coefficient derivatives.

A supersolution is bounded, nonnegative, and continuous on its domain, with the intrinsic classical regularity required in the theorem, and satisfies

$$Yu - A : D_v^2 u \geq 0. \quad (15)$$

The intrinsic Itô identity makes $u(T_s, X_s, V_s)$ a supermartingale up to exit. All stopped comparisons below may first be localized in compact interiors. Nonnegativity, boundedness, and continuity give the stated inequalities on taking the localization limit.

4.1 Near-full cylinders and forward stacks

For an integer $m \geq 1$ define the forward stack

$$\overline{Q}_r^m(t_0, x_0, v_0) = \{(t, x, v) : 0 < t - t_0 \leq mr^2, \quad |v - v_0| < r, \\ |x - x_0 - (t - t_0)v_0| < (m + 2)r^3\}. \quad (16)$$

Lemma 4.1 (Dense-cylinder propagation). *There is a structural $\delta \in (0, 1/4)$ with the following property. For every integer $m \geq 1$ there is $a_m \in (0, 1/2]$ such that*

$$|\{u \geq 1\} \cap Q_r(z_0)| \geq (1 - \delta)|Q_r| \implies u \geq a_m \quad \text{on } \overline{Q}_r^m(z_0), \quad (17)$$

provided the cylinder and the enlarged controlled tubes described in the proof lie in the supersolution domain. The parameter δ is chosen independently of m .

Proof. First take $r = 1$ and $z_0 = 0$. Set

$$\ell = \min\{1/4, (512d\Lambda)^{-1}\}.$$

Start (14) at time $-1/4$ with $|X_0|, |V_0| \leq 1/4$. If $M_s = V_s - V_0$, then Doob's inequality gives

$$\mathbb{E} \sup_{s \leq \ell} |M_s|^2 \leq 8d\Lambda\ell.$$

Therefore the event $\sup_{s \leq \ell} |M_s| \leq 1/4$ has probability at least $3/4$. On this event $|V_s| \leq 1/2$ and $|X_s - X_0| \leq \ell/2$, so the process stays strictly in Q_1 for all $0 \leq s \leq \ell$.

Choose $\delta \in (0, 1/4)$ so small that (13), with horizon 1 and the starting compact just specified, implies

$$|F| \leq \delta|Q_1| \implies \mathbb{E} \int_0^\ell \mathbf{1}_F(T_s, X_s, V_s) ds \leq \ell/4. \quad (18)$$

This choice precedes the choice of m . Let $E = \{u \geq 1\} \cap Q_1$ and let τ be the exit time from Q_1 . The residence estimate and (18) yield

$$\mathbb{E} \int_0^{\ell \wedge \tau} \mathbf{1}_E(T_s, X_s, V_s) ds \geq \ell/2.$$

For each deterministic s , stopped-supermartingale comparison and nonnegativity give

$$u(-1/4, X_0, V_0) \geq \mathbb{E}[u(T_s, X_s, V_s) \mathbf{1}_{\{s < \tau\}}].$$

Integration in $s \in [0, \ell]$ proves

$$u(-1/4, x, v) \geq \frac{1}{2}, \quad |x|, |v| \leq \frac{1}{4}. \quad (19)$$

No first-hitting-time assertion for an arbitrary Borel set is needed.

A point of $\overline{Q}_1^m$ has time in $(0, m]$ and phase coordinates in a fixed compact depending on m . Join it backward to phase $(0, 0)$ at time $-1/4$ by a cubic Hermite path. The duration is in $[1/4, m + 1/4]$. These paths have a common acceleration bound and lie in a fixed enlarged phase box. The controlled-tube lemma, with final coordinate margin smaller than $1/4$, supplies a common probability $p_m > 0$ of staying in this box and landing in the region of (19). Optional sampling then gives $u \geq p_m/2$ at the starting point. Set $a_m = p_m/2$.

Kinetic translation and dilation prove (17) for general r, z_0 . In particular, a point at the immediate beginning of a stack still has a time gap at least $r^2/4$ from the launch slice. No vanishing-duration control is used. $\square$

4.2 Fixed domains and macroscopic cylinders

Fix m , write $a = a_m$, and put

$$T_0 = m + 2, \quad S = [T_0, T_0 + 1] \times B_1 \times B_1. \quad (20)$$

For an observation point $z_* = (t_*, x_*, v_*) \in S$ we use the domain

$$D_{t_*} = (-2, t_*] \times B_R \times B_R. \quad (21)$$

The radius R will be fixed once and for all. Supersolutions need only have intrinsic derivatives below their top time t_* ; comparison at the top follows by one-sided continuity.

Every $Q_r(z_0) \subset Q_1$ has $r \leq 1$, $-1 \leq t_0 \leq 0$, $|x_0|, |v_0| \leq 1$, and stack top time at most m . The normalized paths in Lemma 4.1, after translation and scaling, have coordinates of the form

$$x = x_0 + (t - t_0)v_0 + r^3\tilde{x}, \quad v = v_0 + r\tilde{v},$$

where $\tilde{x}, \tilde{v}$ range over a fixed bounded set depending only on m . Their times range from $t_0 - r^2/4$ to $t_0 + mr^2$, hence lie in $(-2, m]$. Thus one R contains all those tubes, with a positive phase margin, uniformly for $0 < r \leq 1$.

For a second family of paths, consider the phase center of a stack at time $t_0 + r^2/2$:

$$(x_0 + (r^2/2)v_0, v_0). \quad (22)$$

The phase box about this center with radii $r^3/4$ and $r/4$ is inside the stack. From any $z_* \in S$, the backward duration to that time lies in $[1, T_0 + 2]$. Both endpoints belong to fixed phase compacts. The corresponding cubic paths therefore stay in a common bounded phase box independently of r . Increase R once to contain this second family with margin 1.

Lemma 4.2 (Macroscopic density costs a positive value). *For each fixed $r_0 \in (0, 1)$ there is $b(r_0) > 0$ such that every nonnegative supersolution on D_{t_*} satisfies*

$$u(z_*) \geq a b(r_0) \quad (23)$$

if some $Q_r(z_0) \subset Q_1$ with $r \geq r_0$ satisfies the density condition in (17). The domain radius R is independent of r_0 .

Proof. Lemma 4.1 gives $u \geq a$ on the stack. In the second family of cubic paths above, use a coordinate tube radius smaller than $r_0^3/4$. Its terminal box lies in the phase box about (22). The controlled-tube probability has a uniform positive lower bound $b(r_0)$. The tube remains inside D_{t_*} by the choice of R . Optional sampling proves (23). $\square$

Consequently, if a sequence in this fixed supersolution class has $u_n(z_{*,n}) \rightarrow 0$, then for every fixed $r_0 > 0$, all sufficiently large n have the property

$$Q_r \subset Q_1, \quad |\{u_n \geq 1\} \cap Q_r| \geq (1 - \delta)|Q_r| \implies r < r_0. \quad (24)$$

No rate relating r_0 to $u_n(z_{*,n})$ is required.

4.3 A trimmed geometric expansion

Lemma 4.3 (Kinetic set expansion). *Let $E \subset F$ be bounded measurable sets with $E \subset Q_1$. Suppose every cylinder $Q_r \subset Q_1$ with $|E \cap Q_r| \geq (1 - \delta)|Q_r|$ has radius $r < r_0$ and its entire stack $\overline{Q}_r^m$ is contained in F . If $0 < r_0 < 1/4$ and $mr_0^2 \leq 1$, then*

$$|F \cap Q_1| \geq \gamma(|E| - C_d r_0) - C_d m r_0^2, \quad \gamma = \frac{m}{m+1} \left(1 + 8^{-D_{\text{hom}}} \frac{\delta}{2}\right). \quad (25)$$

The constant C_d depends only on dimension.

Proof. Put $\varepsilon = \delta/2$ and trim the reference cylinder to

$$K_{r_0} = \{-1 + r_0^2 < t < -r_0^2, \quad |x| < 1 - 3r_0^2, \quad |v| < 1 - 2r_0\}.$$

Then $|Q_1 \setminus K_{r_0}| \leq C_d r_0$. For $z = (t, x, v) \in K_{r_0}$, consider the midpoint-centered open cylinder

$$C_r(z) = Q_r(t + r^2/2, x + (r^2/2)v, v),$$

with both temporal faces omitted. Equivalently it is defined by

$$|s - t| < r^2/2, \quad |y - x - (s - t)v| < r^3, \quad |w - v| < r.$$

These cylinders are nested in r , contain z in their interior, and are contained in Q_1 for $0 < r \leq r_0$. Indeed their spatial displacement from x is at most $r^2|v|/2 + r^3 < r_0^2$, and the other two containments follow directly from the displayed margins.

We record the needed covering geometry explicitly. If $Q_r(z_1)$ and $Q_s(z_2)$ intersect and $s \leq 2r$, then the latter is contained in

$$Q_{8r}(t_1 + 8r^2, x_1 + 8r^2 v_1, v_1). \quad (26)$$

To verify this, translate z_1 to zero. Intersection implies $-r^2 < t_2 \leq 4r^2$, $|v_2| < 3r$, and $|x_2| < 21r^3$. Points of the second cylinder have time between $-5r^2$ and $4r^2$, velocity norm less than $5r$, and position norm less than $41r^3$. These bounds imply (26). Its volume multiplier is $8^{D_{\text{hom}}}$. A selection by dyadic radius bins gives a countable disjoint subfamily whose enlargements (26) cover any bounded-radius family. At each bin take a maximal disjoint subfamily avoiding earlier selections; each omitted cylinder meets a selected one with at least half its radius. Countability follows since disjoint open cylinders contain distinct rational points.

This covering estimate also proves Lebesgue differentiation for the centered cylinders $C_r(z)$. It gives the usual weak $(1, 1)$ bound for their centered maximal operator. Approximation of an integrable function by continuous compactly supported functions then proves differentiation, since

the Euclidean diameters of $C_r(z)$ tend to zero locally. Thus at almost every $z \in E \cap K_{r_0}$ the density of E in $C_r(z)$ tends to 1 as $r \downarrow 0$. At $r = r_0$ this density is strictly below $1 - \delta$, by the radius assumption. The density is continuous as a function of positive r , because cylinder boundaries have measure zero. Therefore some $r_z \in (0, r_0)$ has

$$|E \cap C_{r_z}(z)| = (1 - \varepsilon)|C_{r_z}(z)|. \quad (27)$$

Let U be the union of all these open cylinders. It is measurable and contains $E \cap K_{r_0}$ up to a null set. Select a disjoint subfamily C_i as above. Its enlargements cover U , so

$$\sum_i |C_i| \geq 8^{-D_{\text{hom}}} |E \cap K_{r_0}|.$$

Using (27) and disjointness gives

$$\begin{aligned} |U \setminus E| &\geq \varepsilon \sum_i |C_i|, \\ |U| &\geq \left(1 + 8^{-D_{\text{hom}}} \varepsilon\right) |E \cap K_{r_0}| \\ &\geq \left(1 + 8^{-D_{\text{hom}}} \varepsilon\right) (|E| - C_d r_0). \end{aligned} \quad (28)$$

Every one of these cylinders satisfies the density assumption for whole-stack inclusion in F .

We next establish the stack-union inequality

$$\left| \bigcup_j \overline{Q_j}^m \right| \geq \frac{m}{m+1} \left| \bigcup_j Q_j \right|. \quad (29)$$

For a finite family, fix v and omit indices with $|v - v_j| \geq r_j$. Straighten $y = x - tv$. The original cylinder section is contained in

$$I_j \times B_{2r_j^3}(x_j - t_j v), \quad I_j = (t_j - r_j^2, t_j),$$

whereas its stack section contains the product with the same spatial ball and time interval $J_j = (t_j, t_j + mr_j^2)$. The first inclusion uses $|(t - t_j)(v - v_j)| < r_j^3$; the second uses its bound mr_j^3 and the stack radius $(m+2)r_j^3$.

For each fixed y , decompose the finite union of the relevant J_j into connected intervals (c_l, d_l) of lengths L_l . If J_j lies in this component, then $mr_j^2 \leq L_l$ and $c_l \leq t_j \leq d_l$, so $I_j \subset (c_l - L_l/m, d_l)$. Summing these enlarged component lengths gives

$$\left| \bigcup_j I_j \right| \leq \left(1 + \frac{1}{m}\right) \left| \bigcup_j J_j \right|.$$

Fubini in y, v proves (29) for a finite family. Monotone convergence proves it for countable families. An arbitrary union of open original cylinders has a countable subcover with the same union; the stacks of this subfamily suffice, avoiding any measurability issue for an arbitrary union of top-closed stacks.

Every cylinder used here has radius less than r_0 and is contained in Q_1 . Its stack lies in

$$(-1, mr_0^2] \times B_{1+Cr_0^2} \times B_1.$$

Indeed $|x| \leq |x_j| + mr_j^2 |v_j| + (m+2)r_j^3$, and its velocity ball is contained in B_1 . For $mr_0^2 \leq 1$, the part of this product outside Q_1 has volume at most $C_d mr_0^2$. Combining this leakage estimate, (28), whole-stack inclusion in F , and (29) proves (25). $\square$

4.4 The critical-volume argument

Proposition 4.4 (Positive-volume growth). *There are structural constants m, R, T_0 as in (20)–(21) such that for every $q \in (0, 1]$ there is a structural number $\nu(q) > 0$ with the following property. Every nonnegative bounded supersolution on D_{t_*} , with $z_* \in S$, satisfies*

$$|\{u \geq 1\} \cap Q_1| \geq q|Q_1| \implies u(z_*) \geq \nu(q). \quad (30)$$

No common upper bound on the supremum of the admissible supersolutions is prescribed.

Proof. Choose m after δ so large that

$$\gamma = \frac{m}{m+1} \left(1 + 8^{-D_{\text{hom}}} \frac{\delta}{2} \right) > 1.$$

Fix the resulting a, R, T_0, S . Define $\nu(q)$ as the infimum of $u(z_*)$ over all smooth elliptic fields, all bounded nonnegative supersolutions on D_{t_*} , and all $z_* \in S$, subject to the volume condition in (30). Thus $0 \leq \nu(q) \leq 1$, by the constant function 1, and ν is nondecreasing. Lemma 4.2, applied to Q_1 itself, gives $\nu(q) > 0$ for $q \geq 1 - \delta$.

Set

$$q_* = \inf\{q \in (0, 1] : \nu(q) > 0\}.$$

Suppose $q_* > 0$. Choose $q < q_*$ so close to q_* that $\gamma q > q_*$. Since $q_* \leq 1 - \delta < 1$, we can choose

$$q_* < q' < \min\{1, \gamma q\}.$$

Choose a fixed $r_0 \in (0, 1/4)$ so small that $mr_0^2 \leq 1$ and

$$\gamma \left(q - \frac{C_d r_0}{|Q_1|} \right) - \frac{C_d m r_0^2}{|Q_1|} \geq q'. \quad (31)$$

As $\nu(q) = 0$, there is a sequence $(u_n, A_n, z_{*,n})$ satisfying the volume condition at q with $u_n(z_{*,n}) \rightarrow 0$. Put

$$E_n = \{u_n \geq 1\} \cap Q_1, \quad F_n = \{u_n \geq a\} \cap D_{t_{*,n}}.$$

These sets are bounded and measurable, and $E_n \subset F_n$. For sufficiently large n , (24) applies at the fixed radius r_0 . Lemma 4.1 puts the entire stack of every dense cylinder in F_n , including the parts outside Q_1 . All these stacks lie below time $m < t_{*,n}$ and inside the fixed phase domain. Lemma 4.3 and (31) imply

$$|F_n \cap Q_1| \geq q'|Q_1|.$$

The function u_n/a belongs to exactly the same supersolution class. The displayed estimate is its threshold-one volume condition. Since $q' > q_*$, monotonicity and the definition of q_* give $\nu(q') > 0$. Hence

$$u_n(z_{*,n})/a \geq \nu(q') > 0,$$

a contradiction. Thus $q_* = 0$, which, by monotonicity, means $\nu(q) > 0$ for every $q > 0$.

The order of choices is m , then q, q' , then a fixed r_0 , and finally sufficiently large n . The positive tube cost $b(r_0)$ may be arbitrarily small as $r_0 \downarrow 0$; no rate is needed. The infimum defining ν ranges over a class specified entirely by d, λ, Λ and the fixed geometry, so its positive values are structural constants. $\square$

4.5 A fixed oscillation contraction

Let $\kappa = \nu(1/2) > 0$. Suppose $0 \leq u \leq 1$ is a solution on $(-2, T_0 + 1] \times B_R \times B_R$. At least one of

$$\{u \geq 1/2\} \cap Q_1, \quad \{u \leq 1/2\} \cap Q_1$$

has volume at least $|Q_1|/2$. In the first case apply Proposition 4.4 to $2u$, obtaining $u \geq \kappa/2$ throughout S . In the second case apply it to $2(1 - u)$, obtaining $u \leq 1 - \kappa/2$ throughout S . The same alternative is selected for every observation point by the fixed past volume. Therefore

$$\text{osc}_S u \leq 1 - \kappa/2. \quad (32)$$

Choose a structural $\rho \in (0, 1/2)$ with

$$\rho^2(T_0 + 3) < 1, \quad \rho^3 R < 1, \quad \rho R < 1.$$

The kinetic transformation

$$(t, x, v) \mapsto (\rho^2(t - (T_0 + 1)), \rho^3 x, \rho v)$$

embeds the large slab into Q_1 with lateral and bottom margins. Its image of S contains $Q_\rho(0)$, apart from the irrelevant choice of the lower temporal face. Normalizing a nonconstant solution by its oscillation and using (32) gives

$$\text{osc}_{Q_\rho(0)} u \leq (1 - \kappa/2) \text{osc}_{Q_1(0)} u. \quad (33)$$

The coefficient is smooth in the argument so far. The following intrinsic residual approximation extends this contraction to the Borel coefficient class in the theorem; kinetic iteration then yields the stated seminorm with its exact symmetric distance.

5 Intrinsic calculus and passage to the original solution class

The preceding geometric argument is uniform over smooth coefficient fields. We now transfer its conclusion to the Borel coefficients and the exact regularity class in Theorem 1.1. In particular, neither separate time derivatives nor separate position derivatives of the solution will be assumed.

Lemma 5.1 (Localized intrinsic Itô formula). *Let $U \subset \mathbb{R}^{1+2d}$ be open. Suppose that f , Yf , and $D_v^2 f$ are continuous on U , with the derivatives existing in the classical intrinsic sense. Let a continuous adapted process satisfy*

$$dT_s = -ds, \quad dX_s = -V_s ds, \quad dV_s = \sigma_s dW_s, \quad \sigma_s \sigma_s^\top = 2B_s, \quad \lambda I \leq B_s \leq \Lambda I.$$

Up to a bounded stopping time during which the spacetime path stays in a fixed compact subset of U , one has

$$df(T_s, X_s, V_s) = (-Yf + B_s : D_v^2 f)(T_s, X_s, V_s) ds + D_v f(T_s, X_s, V_s) \sigma_s dW_s. \quad (34)$$

Here the first velocity derivatives are continuous locally. In particular, a bounded function with $Yf - B : D_v^2 f \geq 0$ is a supermartingale along the corresponding backward diffusion, after compact localization.

Proof. Use the group operation

$$(t, x, v) \circ (s, y, w) = (t + s, x + y + sv, v + w), \quad (s, y, w)^{-1} = (-s, -y + sw, -w).$$

Lebesgue measure is invariant under left and right translations, and $Y, \partial_{v_1}, \dots, \partial_{v_d}$ are left invariant. Choose a nonnegative compactly supported smooth approximate identity ψ_ε at the identity. On a compact neighborhood of the stopped path, define the left smoothing

$$f_\varepsilon(z) = \int \psi_\varepsilon(\zeta) f(\zeta^{-1} \circ z) d\zeta.$$

One can first multiply f by a smooth cutoff equal to one on a larger compact neighborhood. For sufficiently small ε , that cutoff has no effect on this formula or its derivatives at the points under consideration. A change of variables transfers all derivatives to the smooth compact kernel, so f_ε is smooth.

If $\zeta = (s, y, w)$, then

$$\zeta^{-1} \circ z = (t - s, x - y - (t - s)w, v - w).$$

Consequently left smoothing commutes with the displayed left invariant derivatives:

$$Y f_\varepsilon = \psi_\varepsilon * (Y f), \quad D_v^2 f_\varepsilon = \psi_\varepsilon * (D_v^2 f).$$

These identities first hold for distributional derivatives. The given classical derivatives agree with them by the fundamental theorem of calculus along the flow of Y and along velocity lines. Hence $f_\varepsilon, Y f_\varepsilon, D_v^2 f_\varepsilon$ converge uniformly on the compact neighborhood to $f, Y f, D_v^2 f$.

For completeness, for a fixed sufficiently small positive h the one-dimensional Taylor formula on a velocity line gives

$$\partial_{v_i} f(z) = \frac{f(z + h e_{v_i}) - f(z)}{h} - \frac{1}{h} \int_0^h (h - s) \partial_{v_i v_i} f(z + s e_{v_i}) ds.$$

The right side is continuous locally. Applying the same formula to $f_\varepsilon - f$ proves uniform convergence of $D_v f_\varepsilon$ on a slightly smaller compact set.

The ordinary Itô formula applies to f_ε . All drift integrands converge uniformly. The stochastic integrals converge in L^2 by the Itô isometry, boundedness of B_s , and uniform convergence of $D_v f_\varepsilon$. This proves (34). The drift is nonpositive for a supersolution, giving the final assertion. $\square$

The same argument permits a closed top time face without assuming derivative bounds there. More precisely, let a bounded f be continuous on a slab up to its top, with the hypotheses of Lemma 5.1 strictly below that top. Start the backward process at the top and stop it inside a compact set whose lower and lateral faces lie strictly inside the slab. On each interval $[a, b]$ with $a > 0$, the stopped process is a supermartingale by the lemma; paths stopped before a are constant. As $a \downarrow 0$, continuity of the path and of f gives its initial value, and boundedness permits passage through conditional expectations. Thus the supermartingale comparison extends to the initial top point. For an exact solution the identical reasoning gives a martingale. This argument uses cancellation of the drift only at positive elapsed times; it does not require the separate terms $Y f$ and $D_v^2 f$ to be integrable up to the top face.

Lemma 5.2 (One smooth field with uniformly small equation residual). *Suppose $u, Yu, D_v^2 u$ are continuous on an open set U and $Yu = A : D_v^2 u$ there, where A is Borel and $\lambda I \leq A \leq \Lambda I$. For every compact $K \Subset U$ there are smooth fields A_n on all of $\mathbb{R}^{1+2d}$, satisfying the same ellipticity bounds, such that*

$$\eta_n := \sup_K |Yu - A_n : D_v^2 u| \longrightarrow 0. \quad (35)$$

Proof. Extend A outside U by λI and convolve the resulting bounded Borel field with a nonnegative Euclidean smooth approximate identity ϕ_n . The fields $A_n = A * \phi_n$ are globally smooth. Since the matrix interval $[\lambda I, \Lambda I]$ is convex, they retain the same ellipticity bounds.

Write $H = D_v^2 u$ and $g = Yu$. For $z \in K$ and all sufficiently large n , the convolution only uses points of a fixed compact neighborhood of K inside U . The equation gives the exact identity

$$A_n(z) : H(z) - g(z) = \int \phi_n(z - y) [g(y) - g(z) + A(y) : (H(z) - H(y))] dy.$$

The absolute value is bounded by the modulus of continuity of g plus $d\Lambda$ times the operator-norm modulus of continuity of H , on that neighborhood and at the radius of the convolution. Both tend to zero uniformly on K . $\square$

The approximation in Lemma 5.2 concerns the equation residual. It requires no uniform approximation of the Borel field itself. The same A_n works throughout K .

Proposition 5.3 (Oscillation contraction in the original class). *There exist structural constants $\rho \in (0, 1)$ and $q \in (0, 1)$ such that every solution in the precise class of Theorem 1.1 satisfies*

$$\text{osc}_{Q_{\rho r}(z_0)} u \leq q \text{osc}_{Q_r(z_0)} u. \quad (36)$$

Proof. We use the structural smooth-coefficient growth conclusion of Theorem 4.4, in its supersolution form. To state precisely the input used here, let $T_0 > 1$, $R > 1$, and $\kappa \in (0, 1]$ be its structural constants. For every

$$z_* = (t_*, x_*, v_*) \in [T_0, T_0 + 1] \times B_1 \times B_1, \quad D_{t_*} = (-2, t_*] \times B_R \times B_R,$$

a bounded nonnegative continuous intrinsic supersolution f for any smooth elliptic field on D_{t_*} obeys

$$|\{f \geq 1\} \cap Q_1| \geq \frac{1}{2}|Q_1| \implies f(z_*) \geq \kappa. \quad (37)$$

The derivatives need only exist continuously below the top time. The variable observation top in this statement is useful for the following transfer.

By translation and dilation it suffices initially to consider $Q_1(0)$. Normalize a nonconstant solution by its oscillation so that $0 \leq u \leq 1$ there. Choose a structural $a \in (0, 1/2)$ such that

$$a^2(T_0 + 3) < 1, \quad a^3 R < 1, \quad aR < 1,$$

and put

$$w(t, x, v) = u(a^2(t - T_0 - 1), a^3 x, av).$$

This defines w continuously on the closure of $(-2, T_0 + 1] \times B_R \times B_R$. All its lower and lateral faces, and every compact set below its top, map strictly inside the original cylinder. It has the required intrinsic derivatives below $T_0 + 1$, satisfies an equation with a Borel coefficient in the same ellipticity interval, and $0 \leq w \leq 1$.

At least one of the sets

$$\{w \geq \frac{1}{2}\} \cap Q_1, \quad \{w \leq \frac{1}{2}\} \cap Q_1$$

has measure at least $|Q_1|/2$. Fix the alternative once, independently of the observation point. Suppose first that the first set does. Fix z_* as in (37) with $t_* < T_0 + 1$. The closure of D_{t_*} has a compact neighborhood below the top on which w and its intrinsic derivatives have the stated

continuity. Apply Lemma 5.2 on that neighborhood to obtain one smooth field A_n and residual bound $\eta_n \rightarrow 0$. On D_{t_*} define

$$f_n(t, x, v) = 2w(t, x, v) + 2\eta_n(t + 2).$$

It is bounded and nonnegative, and

$$Y f_n - A_n : D_v^2 f_n = 2(Y w - A_n : D_v^2 w + \eta_n) \geq 0.$$

The selected half-volume set is contained in $\{f_n \geq 1\}$. Therefore (37) gives

$$w(z_*) \geq \frac{\kappa}{2} - \eta_n(t_* + 2).$$

Letting $n \rightarrow \infty$ proves $w(z_*) \geq \kappa/2$. The other alternative is handled by $f_n = 2(1 - w) + 2\eta_n(t + 2)$ and gives $w(z_*) \leq 1 - \kappa/2$. Because the same alternative was fixed on the past cylinder, it holds for all observation points with $t_* < T_0 + 1$. Continuity extends it to $t_* = T_0 + 1$. Consequently

$$\text{osc}_{[T_0, T_0+1] \times B_1 \times B_1} w \leq 1 - \kappa/2.$$

The image of this observation set is $[-a^2, 0] \times B_{a^3} \times B_a$, which contains $Q_a(0)$. This proves the unit-scale contraction with $\rho = a$ and $q = 1 - \kappa/2$. Constant solutions are immediate. The exact scaling proved below gives (36) at every center and radius. $\square$

6 The exact kinetic distance and the Hölder seminorm

We finish with the geometric passage from (36) to the exact seminorm in the question. This also verifies all radius, center, and top-face quantifiers. Define

$$\Phi_{r,z_0}(t, x, v) = (t_0 + r^2 t, x_0 + r^3 x + r^2 t v_0, v_0 + r v).$$

This map takes $Q_1(0)$ exactly onto $Q_r(z_0)$ and takes the open time interiors to one another. The chain rule along the transport flow and velocity lines gives

$$Y(u \circ \Phi_{r,z_0}) = r^2(Yu) \circ \Phi_{r,z_0}, \quad D_v^2(u \circ \Phi_{r,z_0}) = r^2(D_v^2 u) \circ \Phi_{r,z_0}.$$

Thus the equation and its precise regularity class are preserved, with unchanged ellipticity constants. Substituting $w = v_0 + r w$ in the defining minimum of the distance gives

$$d_{\text{kin}}(\Phi_{r,z_0}(z), \Phi_{r,z_0}(z')) = r d_{\text{kin}}(z, z'). \quad (38)$$

Completion of the proof of Theorem 1.1. The objective minimized in d_{kin} is a continuous coercive function of w , so the minimum is attained. For two distinct points $z_i = (t_i, x_i, v_i)$, let $\delta = d_{\text{kin}}(z_1, z_2) > 0$ and choose a minimizing w . The positivity of δ follows directly from the definition: zero distance forces equality of all coordinates. We have

$$|t_1 - t_2| \leq \delta^2, \quad |v_i - w| \leq \delta, \quad |x_1 - x_2 - (t_1 - t_2)w| \leq 2\delta^3.$$

Set

$$t_c = \max(t_1, t_2), \quad x_c = \frac{1}{2} \sum_{i=1}^2 [x_i - (t_i - t_c)w], \quad z_c = (t_c, x_c, w).$$

For each i , the spatial error relative to the characteristic through z_c is at most δ^3 , its velocity error is at most δ , and its backward time displacement is at most δ^2 . In particular,

$$z_1, z_2 \in Q_{2\delta}(z_c). \quad (39)$$

The strict lateral and lower-time inequalities in this inclusion follow from the factor 2.

Assume now that $z_1, z_2 \in Q_{1/2}(0)$ and $\delta \leq 1/32$. Since $|v_i| < 1/2$, $|x_i| < 1/8$, and $t_c > -1/4$,

$$|w| < \frac{17}{32}, \quad |x_c| \leq \frac{1}{8} + \frac{17}{64}\delta^2 < \frac{1}{7}.$$

Every point of $Q_{1/8}(z_c)$ has velocity norm below $21/32$, position norm below

$$\frac{1}{7} + \frac{1}{64} \frac{17}{32} + \frac{1}{512} < 1,$$

and time above $-1/4 - 1/64$ and at most 0. Hence

$$Q_{1/8}(z_c) \subset Q_1(0). \quad (40)$$

Let ρ, q be the structural constants in Proposition 5.3, and put

$$\beta = \frac{\log q}{\log \rho} > 0, \quad \alpha = \min\{\frac{1}{2}, \frac{\beta}{2}\} \in (0, 1).$$

Iterating (36) on the concentric cylinders in (40) yields

$$\text{osc}_{Q_{\rho^k/8}(z_c)} u \leq q^k \text{osc}_{Q_1(0)} u, \quad k \geq 0.$$

Choose the largest integer $k \geq 0$ for which $\rho^k/8 \geq 2\delta$. Such an integer exists since $\delta \leq 1/32$, and its maximality gives $\rho^k < 16\delta/\rho$. By (39),

$$|u(z_1) - u(z_2)| \leq q^k \text{osc}_{Q_1} u \leq \left(\frac{16\delta}{\rho}\right)^\beta \text{osc}_{Q_1} u \leq C\delta^\alpha \text{osc}_{Q_1} u.$$

When $\delta > 1/32$, the same bound follows directly from $|u(z_1) - u(z_2)| \leq \text{osc}_{Q_1} u$, after increasing C . All comparison cylinders may have a top face at time zero; Proposition 5.3 already includes that face by continuity. No derivative at the top has been used. Taking the supremum over pairs proves

$$[u]_{C_{\text{kin}}^\alpha(Q_{1/2}(0))} \leq C \text{osc}_{Q_1(0)} u.$$

Finally apply this inequality to $u \circ \Phi_{r,z_0}$ and use (38). This gives precisely

$$r^\alpha [u]_{C_{\text{kin}}^\alpha(Q_{r/2}(z_0))} \leq C \text{osc}_{Q_r(z_0)} u,$$

with α and C depending only on d, λ, Λ . □

Verification and scope

The accompanying audit record identifies the frozen mathematical sources and their independent whole-proof and interface audits. These are mathematical audits by language-model agents, not formal proof certificates or external peer review. The proof above, rather than an audit verdict, is the mathematical content of this manuscript. The constants may be extremely small or large as functions of the ellipticity ratio and dimension. The occupation exponent is explicit; the final Hölder exponent uses a positive infimum established by the critical-volume argument and is not given as an explicit numerical formula.

References

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